

Using AWS Config

(LAB-M07-01)

Overview

In this lab, you set up AWS Config to monitor your Amazon Simple Storage Service (Amazon S3) buckets for security violations. Your company wants to ensure that Amazon S3 buckets are publicly accessible. To accomplish this, use Amazon CloudWatch Events in concert with AWS Config to monitor the buckets in near-real time. If a violation is found, CloudWatch will trigger an AWS Lambda function to bring the resource back into compliance.

Objectives

After completing this lab, you will be able to:

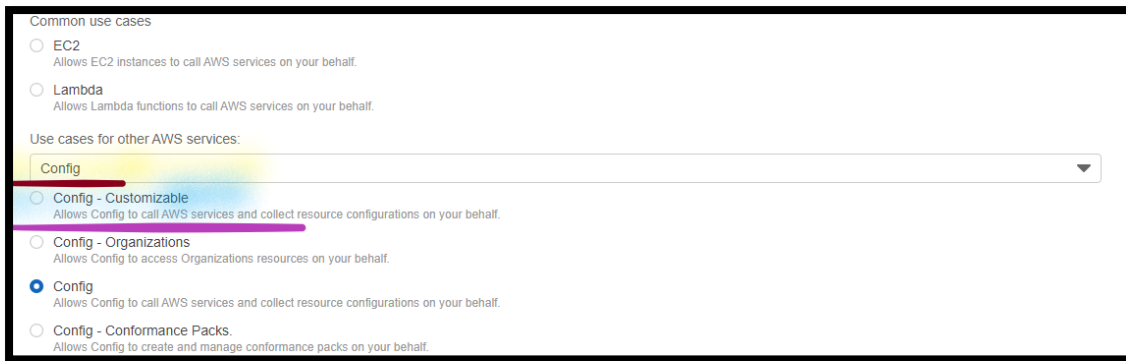
- Set up AWS Config to monitor Amazon S3 bucket access control lists (ACLs) and policies for compliance violations
- Create and configure a CloudWatch Events rule that is triggered by AWS Config and subsequently triggers a Lambda function
- Explain how to use a Lambda function to correct Amazon S3 bucket ACL policies and send notifications about out-of-compliance policies

Task 1: Create IAM Role

In this task you will create IAM roles for AWS Config and Lambda function.

Step 1: Create IAM Roles for AWS Config

1. In the **AWS Management Console**, on the **Services** menu Search and Select **IAM**.
2. Select **Roles**, click on **Create role**.
 - a. In the **Select trusted entity** section:
 - i. **Trusted entity type**: Select **AWS service**.
 - ii. **Common use cases**: Dropdown and select **Config**.
 - a) Select **Config - customizable**.



Common use cases

☐ EC2
Allows EC2 instances to call AWS services on your behalf.

☐ Lambda
Allows Lambda functions to call AWS services on your behalf.

Use cases for other AWS services:

☒ Config
Allows Config to call AWS services and collect resource configurations on your behalf.

☐ Config - Customizable
Allows Config to call AWS services and collect resource configurations on your behalf.

☐ Config - Organizations
Allows Config to access Organizations resources on your behalf.

☒ Config
Allows Config to call AWS services and collect resource configurations on your behalf.

☐ Config - Conformance Packs
Allows Config to create and manage conformance packs on your behalf.

iii. Select **Next**.

b. In the **Add Permissions** section:

Note: Leave the details as default.

i. Select **Next**.

c. In the **Name, review, and create** section:

i. **Role Name:** Write **ConfigRole**.

ii. Click **Create role**.

Note: **Wait**, till Role **ConfigRole** gets created.

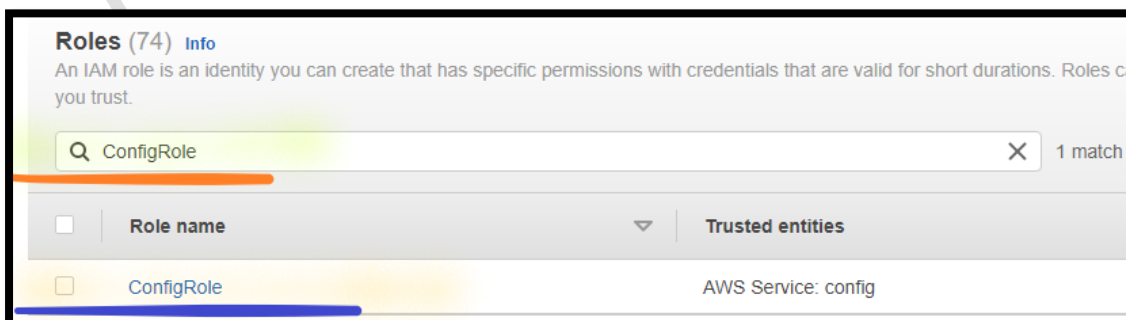
Configure AWS Config Roles

3. **From** the **IAM** console:

4. Select **Roles**.

a. in the **Search box**, type **ConfigRole**, press **Enter**.

b. Open the **ConfigRole**.



Roles (74) [Info](#)

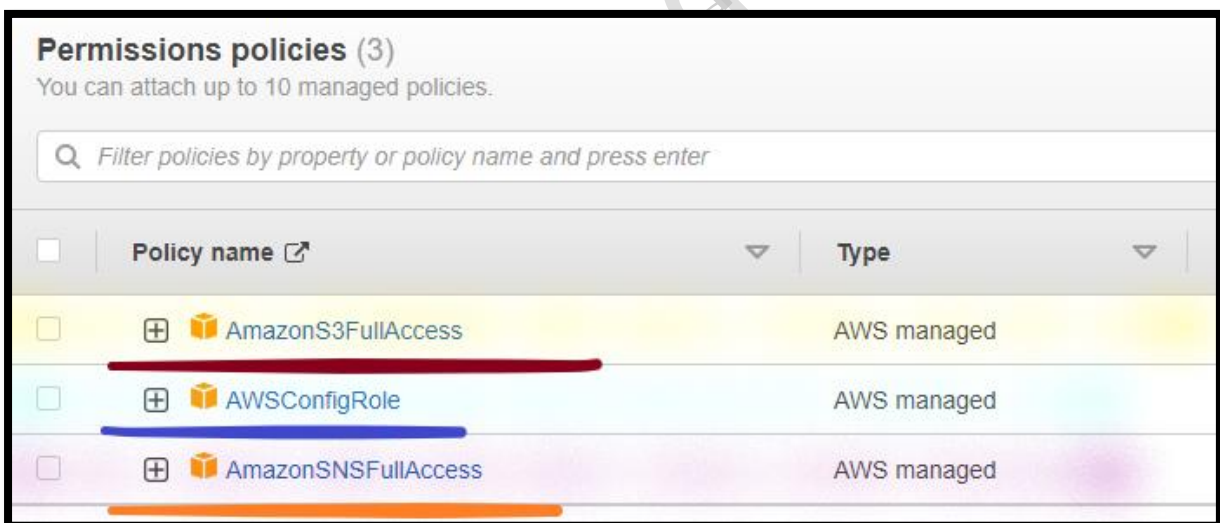
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assigned to users, groups, or other roles that you trust.

Q ConfigRole X 1 match

Role name	Trusted entities
ConfigRole	AWS Service: config

- c. Select **Add permission**.
- d. Select **Attach policies**.
 - i. in the **Search box**, type **AmazonSNSFullAccess**, press **Enter**.
 - a) Select **AmazonSNSFullAccess**.
 - b) Select **Clear Filter**.
 - ii. in the **Search box**, type **AmazonS3FullAccess**, press **Enter**.
 - a) Select **AmazonS3FullAccess**.
 - iii. Select **Attach policies**.

Note: In the **Permission policy summary**, you can see the three policies **AWSConfigRole** and **AmazonSNSFullAccess** and **AmazonS3FullAccess**.



Step 2: Create IAM Role for Lambda Function

5. **From** the **IAM** console:
6. Select **Roles**, click on **Create role**.
7. Select **Lambda**, under a **use case**.
 - a. Select **Next: Permissions**.
 - i. Search and select **AmazonSNSFullAccess**.

- ii. Search and select **CloudWatchLogsFullAccess**.
- iii. Search and select **AmazonS3FullAccess**.
- iv. Search and select **AWSLambdaBasicExecutionRole**.
- b. Select **Next: Tags**.
- c. Select **Next: Review**.
 - i. **Role name:** Write **LambdaConfigRole**.
- d. Click **Create role**.

Note: You get the message, the role **LambdaConfigRole** been created.

Task 2: Create SNS Topic

In this task you will create SNS topic for email notification.



Step 1: Create Topic

- 8. In the **AWS Management Console**, on the **Services** menu, click **Simple Notification Service**.
- 9. Choose the **US East (N. Virginia)** region list to the right of your account information on the navigation bar.

10. Click **Start with overview**.

11. Click **Topics**.

a. Click **Create topic** and configure:

i. **Type**: Select **Standard**.

ii. **Topic name**: Write **ConfigSNSTopic**.

iii. **Display name**: Write **ConfigSNSTopic**.

b. Click **Create topic**.

12. **Copy** the **Topic ARN** in a **Notepad**.

Step 2: Subscribe the Topic

13. From the **Simple Notification Service** console:

14. Select the **Topics**

15. Open the **ConfigSNSTopic**.

a. Click **Create subscription** and configure:

i. **Protocol:** Dropdown and select **Email**.

ii. **Endpoint:** Write **your email id**.

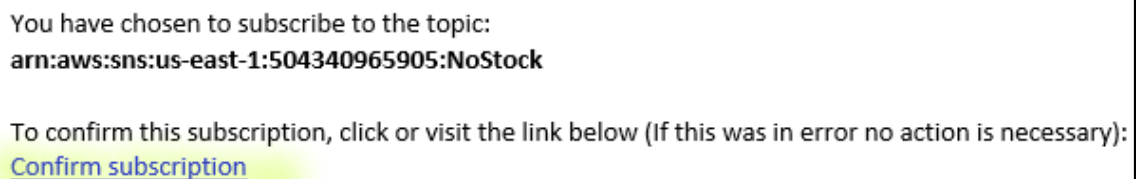


The screenshot shows the 'Create subscription' form in the Amazon SNS console. It has three main sections: 'Topic ARN' with a text input field containing 'arn:aws:sns:us-east-1:504340965905:ConfigSNSTopic'; 'Protocol' with a dropdown menu set to 'Email'; and 'Endpoint' with a text input field containing a partially obscured email address ending in '@gmail.com'. Each section has a small description below it.

b. Click **Create subscription**.

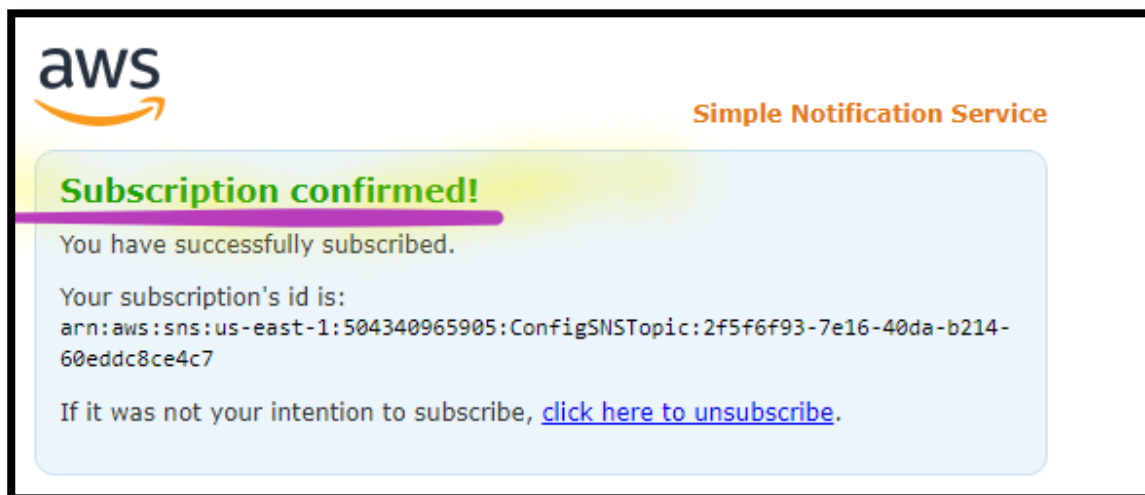
Note: After creating an Email subscription, a confirmation email will be sent to you. Open the email and click the *Confirm subscription* link.

16. You will also **receive email** for subscription confirmation. Click on **Confirm subscription**.



The screenshot shows an email confirmation message. It states: 'You have chosen to subscribe to the topic: arn:aws:sns:us-east-1:504340965905:NoStock'. Below this, it says: 'To confirm this subscription, click or visit the link below (If this was in error no action is necessary):' followed by a blue hyperlink labeled 'Confirm subscription'.

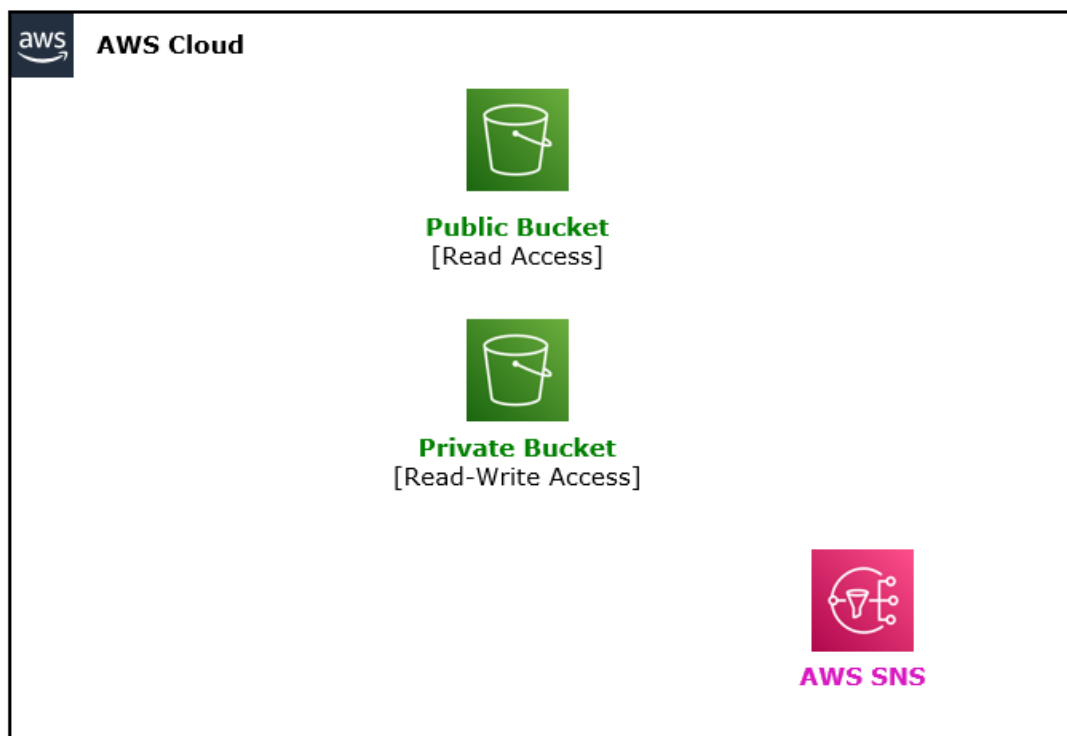
Note: You can see the message **Subscription Confirmed**.



Task 3: Create AWS S3 Bucket

In this task you create an S3 bucket for AWS Configuration and privatebucket and publicbucket. The publicbucket is supposed to be publicly accessible, while the privatebucket is not supposed to have public access.

A custom tag, CanBePublic, indicates if the bucket should or should not be publicly accessible.



Step 1: Create First Bucket

17. In the AWS Management Console, on the **Services** menu, click **S3**.

18. Click **Create bucket**.

a. **Bucket name:** Write **configbucket-123**.

Note: Replace **123** with a random number to make bucket name unique.

b. **Region:** Dropdown and Select **US East (N. Virginia)**.

c. **Object Ownership:** Select **ACL Enabled**.

i. Select the **Object Writer**.

Object Ownership Info

Control ownership of objects written to this bucket from other AWS accounts and granted using access control lists (ACLs). Object ownership determines who can specify access to objects.

☐ ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

☐ Bucket owner preferred
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.

☒ Object writer
The object writer remains the object owner.

d. **Block all public access:** **Deselect** (remove the checkmark) the option.

i. **Select** the *I acknowledge that the current settings might result in this bucket and the objects within becoming public.*

☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

 **Turning off block all public access might result in this bucket and the objects within becoming public**
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Note: Leave other details as default.

- e. Select the **Create bucket**.

Step 2: Create Second Bucket

19. From **S3** console:

20. Click **Create bucket**.

- a. **Bucket name:** Write **publicbucket-123**.

Note: Replace **123** with a random number to make bucket name unique.

- b. **Region:** Dropdown and Select **US East (N. Virginia)**.

- c. **Object Ownership:** Select **ACL Enabled**.

- i. Select the **Object Writer**.

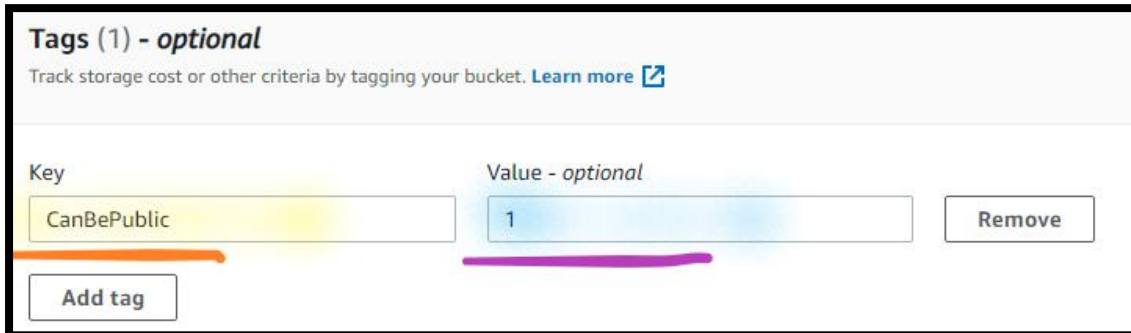
- d. **Block all public access:** **Deselect** (remove the checkmark) the option.

- i. **Select** the ***I acknowledge that the current settings might result in this bucket and the objects within becoming public.***

e. **Tag:** Select **Add Tag**:

i. **Name:** Write **CanBePublic**.

ii. **Value:** Write **1**.



Tags (1) - optional
Track storage cost or other criteria by tagging your bucket. [Learn more](#)

Key	Value - optional	
CanBePublic	1	Remove

Add tag

Note: Leave other details as default.

Note: There are several tags applied to the bucket. Notice the CanBePublic tag has a value of 1. This indicates that the bucket can be public.

f. Select the **Create bucket**.

Configure ACL [Read] for public bucket

21. From **S3** console:

22. Select **Buckets**.

a. Open **publicbucket-123**.

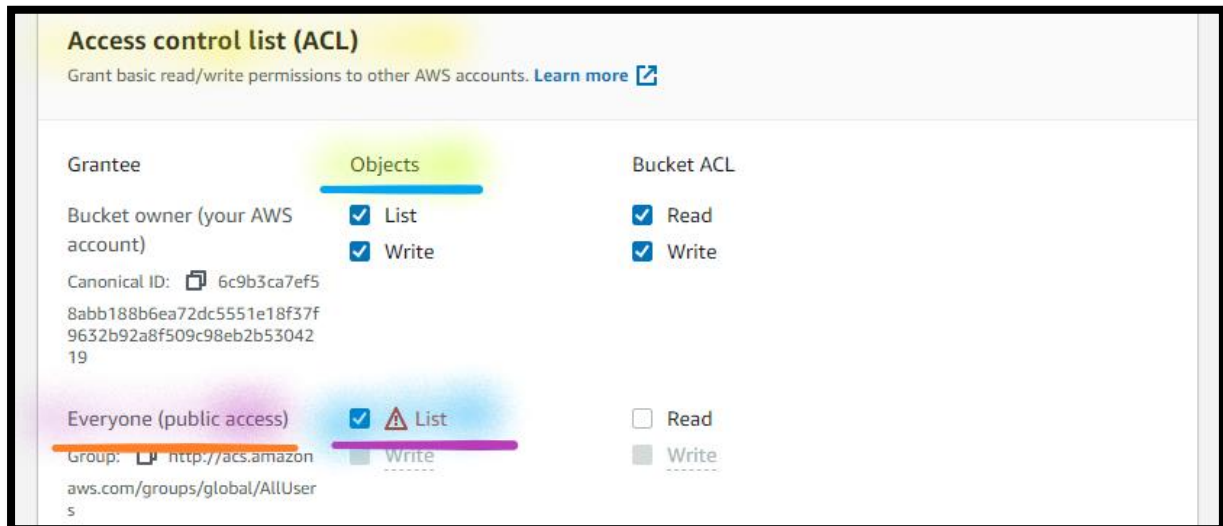
b. Select **Permissions**.



publicbucket-123 Info

Objects | Properties | **Permissions** | Metrics | Management | Access Points

- c. **Access control list (ACL):** Select **Edit**.
- Everyone (public access):** Select **List** under **Objects**.
 - Select **I understand the effects of these changes on my objects and buckets**.



Note: You have to enable the Public Read access to the S3 Objects.

- Select **Save changes**.

Step 3: Create Third Bucket

23. From **S3** console:

24. Click **Create bucket**.

- Bucket name:** Write **privatebucket-123**.

Note: Replace **123** with a random number to make bucket name unique.

- Region:** Dropdown and Select **US East (N. Virginia)**.
- Object Ownership:** Select **ACL Enabled**.
 - Select the **Object Writer**.
- Block all public access:** **Deselect** (remove the checkmark) the option.

- i. **Select** the ***I acknowledge that the current settings might result in this bucket and the objects within becoming public.***

e. **Tag:** Select **Add Tag**:

- i. **Name:** Write **CanBePublic**.
- ii. **Value:** Write **0**.

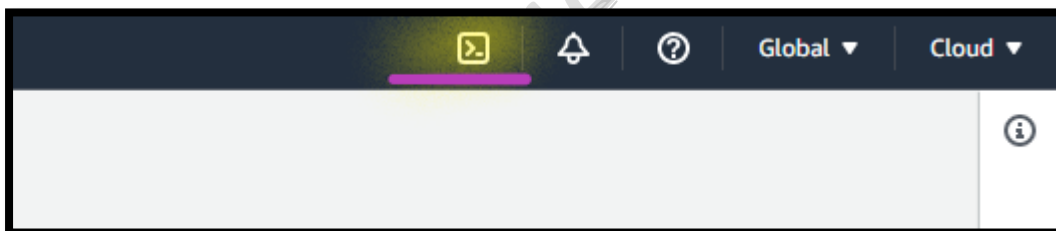
Note: Leave other details as default.

Note: Notice the CanBePublic tag has a value of 0. This indicates that the bucket should not be public.

- f. Select the **Create bucket**.

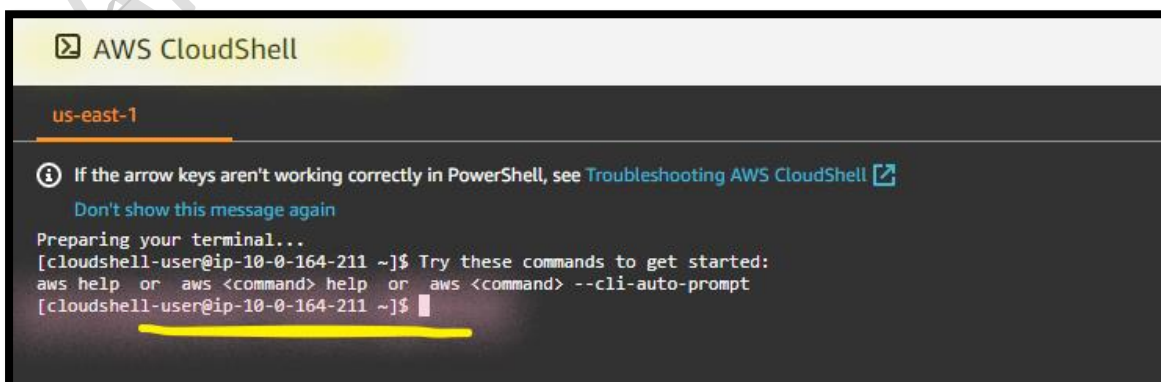
Configure ACL [Read-Write] for private bucket

25. Select **CloudShell**.



26. In the **Welcome to CloudShell**, select **Close**.

Note: **Wait**, till you can see the **Prompt**.

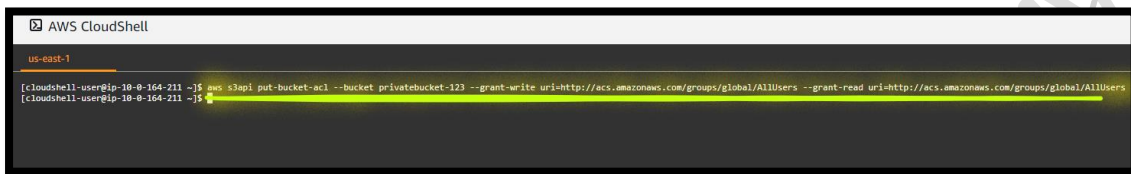


27. From **CloudShell**:

- a. **Execute** the *below command*.

```
aws s3api put-bucket-acl --bucket privatebucket-123 --grant-write  
uri=http://acs.amazonaws.com/groups/global/AllUsers --grant-read  
uri=http://acs.amazonaws.com/groups/global/AllUsers
```

Note: Replace the **privatebucket-123** with **your Bucket name**.



```
AWS CloudShell  
us-east-1  
[cloudshell-user@ip-10-0-164-211 ~]$ aws s3api put-bucket-acl --bucket privatebucket-123 --grant-write  
uri=http://acs.amazonaws.com/groups/global/AllUsers --grant-read uri=http://acs.amazonaws.com/groups/global/AllUsers  
[cloudshell-user@ip-10-0-164-211 ~]$
```

28. From **S3** console:

29. Select **Buckets**.

- a. Open **privatebucket-123**.
b. Select **Permissions**.
c. **Access control list (ACL)**: Select **Edit**.

Note: You can see **Everyone permission** as **List** and **Write**.

i. **Bucket owner (your AWS account):**

- a) Select **List** and **Write** under **Objects**.
b) Select **Read** and **Write** under **Bucket ACL**.

ii. **Select** *I understand the effects of these changes on my objects and buckets*.

Access control list (ACL)

Grant basic read/write permissions to other AWS accounts. [Learn more](#)

Grantee	Objects	Bucket ACL
Bucket owner (your AWS account) Canonical ID: 6c9b3ca7ef58abb188b6ea72dc5551e18f37f9632b92a8f509c98eb2b5304219	☒ List ☒ Write	☒ Read ☒ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☒ List ☒ Write	☐ Read ☐ Write

Note: You have to enable the Public Read and Write access to the S3 Objects.

iii. Select **Save changes**.

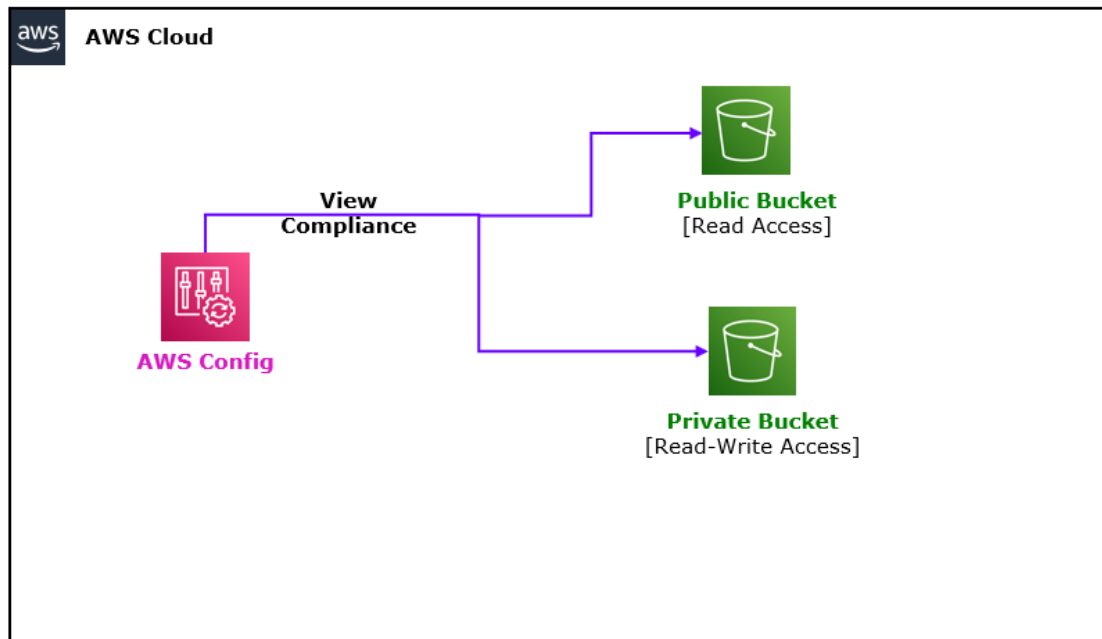
Task 4: Configure AWS Config

In this task, you are configuring the AWS Config who monitor the resources in the account and will record if a policy change has been applied.

You want to ensure that those changes comply with company rules. When a change does not comply, you want to correct the policy to ensure compliance.

AWS Config will send a message to CloudWatch Events when there is a resource that is not compliant with your rule.

To do this, you will create two rules. One rule is to block public read access (*S3ProhibitPublicReadAccess*), and the other is to block public write access (*S3ProhibitPublicWriteAccess*).



Step 1: Configure AWS Config

30. In the **AWS Management Console**, on the **Services** menu, click **Config**.

31. **Go to the left**, select **Setup AWS Config** and **Configure**:

a. In the **General settings** section:

- i. **Resource types to record**: Select **Record specific resource types**.
- ii. **Resource category**: Dropdown and select **AWS resources**.
- iii. **Resource type**: Dropdown and select **AWS S3 Bucket**.
- iv. **AWS Config role**: Select **Choose a role from your account**.
 - a) **Existing roles**: Dropdown and select **ConfigRole**.

Resource types to record

☐ Record all resources supported in this region ☒ Record specific resource types

To learn more, see [Supported Resource Types](#)

Resource category: AWS resources

Resource type: Multiple Selected

AWS S3 Bucket X

AWS Config role

☐ Use an existing AWS Config service-linked role ☒ Choose a role from your account

Existing roles: ConfigRole

b. In the **Delivery method** section:

- i. **Amazon S3 bucket:** Select **Choose a bucket from your account**.
 - a) **S3 bucket name:** Dropdown and select **configbucket-123**.
- ii. **Amazon SNS topic:** **Enable** Stream configuration changes and notifications to an Amazon SNS topic.
 - b) Select **Choose a topic from your account**.
 - c) **SNS topic name:** Dropdown and select **ConfigSNSTopic**.

Amazon S3 bucket

☐ Create a bucket ☒ Choose a bucket from your account ☐ Choose a bucket from another account

Ensure appropriate permissions are available in this S3 bucket's policy. [Learn more](#)

S3 bucket name: configbucket-123

Prefix (optional): /AWSLogs/504340965905/Config/us-east-1

Amazon SNS topic

☒ Stream configuration changes and notifications to an Amazon SNS topic. If you choose email as the notification endpoint for your SNS topic, this can cause a high volume of email. [Learn more](#)

☐ Create a topic ☒ Choose a topic from your account ☐ Choose a topic from another account

Ensure appropriate permissions are available in this SNS topic's policy. [Learn more](#)

SNS topic name: ConfigSNSTopic

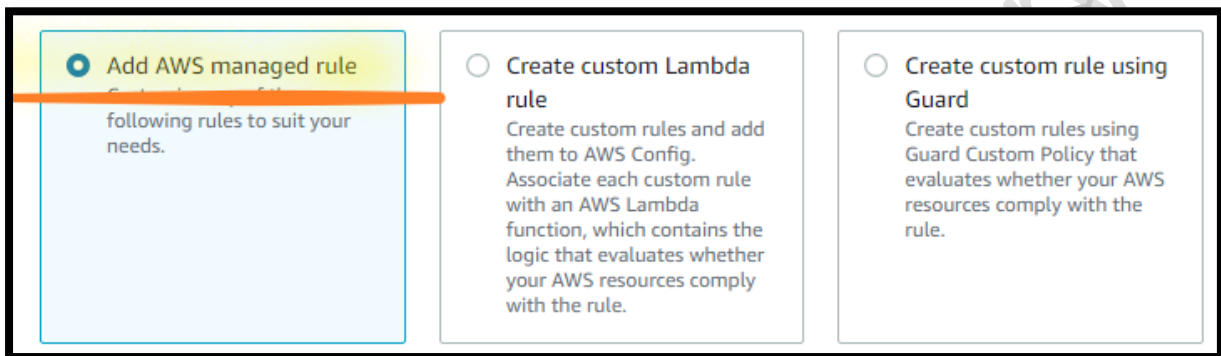
c. Select **Save**.

Step 2: Create First AWS Config Rule

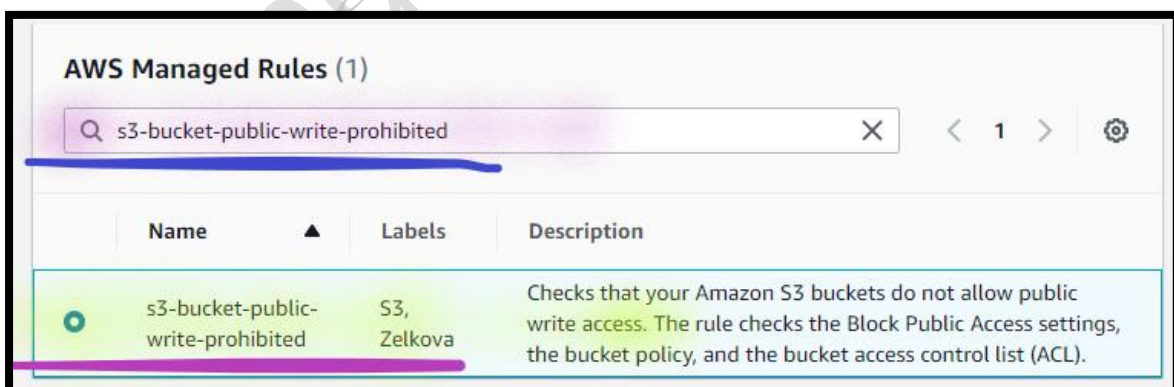
32. From the **Config** console:

33. **Go to the left**, select **Rules**.

- a. Select **Add Rule**.
- b. In the **Select rule type** section:
 - i. Select **Add AWS managed rule**.



- c. In the **AWS Managed Rules** section:
 - i. In the **Search box**, type **s3-bucket-public-write-prohibited**.
 - ii. Select **s3-bucket-public-write-prohibited**.
 - iii. Select **Next**.



Note: s3-bucket-public-write-prohibited

The rule checks the Block Public Access settings, the bucket policy, and the bucket access control list (ACL).

The rule is compliant when both of the following are true:

1. The Block Public Access setting restricts public policies or the bucket policy does not allow public write access.
2. The Block Public Access setting restricts public ACLs or the bucket ACL does not allow public write access.

- d. In the **Config rule** page:

Note: Leave other details as default.

- i. Select the **Next**.
- e. In the **Review and create** page:
- i. Select the **Add rule**.

Step 3: Create Second AWS Config Rule

34. From the **Config** console:

35. **Go to the left**, select **Rules**.

- a. Select **Add Rule**.
- b. In the **Select rule type** section:
 - i. Select **Add AWS managed rule**.
- c. In the **AWS Managed Rules** section:
 - i. In the **Search box**, type **s3-bucket-public-read-prohibited**.
 - ii. Select **s3-bucket-public-read-prohibited**.
 - iii. Select **Next**.
- d. In the **Config rule** page:

Note: Leave other details as default.

Note: s3-bucket-public-read-prohibited

Checks if your Amazon S3 buckets do not allow public read access. The rule checks the Block Public Access settings, the bucket policy, and the bucket access control list (ACL).

The rule is compliant when both of the following are true:

1. The Block Public Access setting restricts public policies or the bucket policy does not allow public read access.
2. The Block Public Access setting restricts public ACLs or the bucket ACL does not allow public read access.

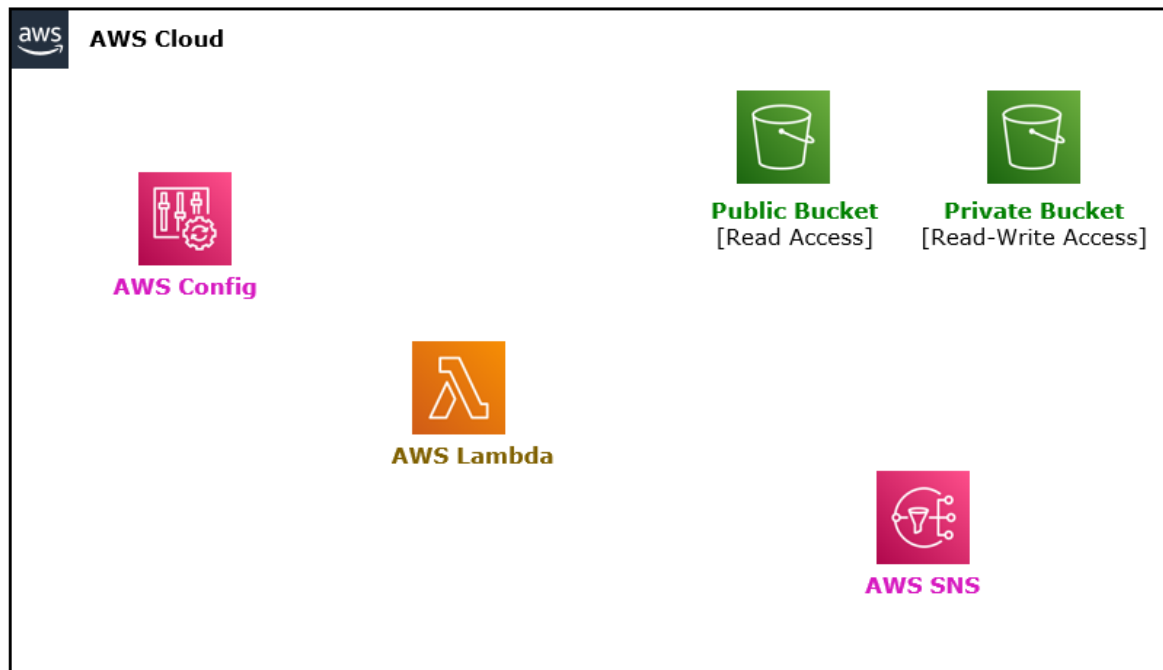
- i. Select the **Next**.
- e. In the **Review and create** page:
 - i. Select the **Add rule**.

Note: You can see the two **Config rules** with **Non-Compliant Resources**.

Rules				View details	Edit rule
Any status ▼					
Name	Remediation action	Type	Compliance		
☐ s3-bucket-public-write-prohibited	Not set	AWS managed	⚠ 1 Noncompliant resource(s)		
☐ s3-bucket-public-read-prohibited	Not set	AWS managed	⚠ 2 Noncompliant resource(s)		

Task 5: Create AWS Lambda Function

The Lambda function will scan all your buckets and look for the custom tag CanBePublic. If any bucket is assigned public access and does not have the tag CanBePublic set to 1, the Lambda function will treat it as a policy violation. When this happens, it will set the bucket ACL to private and send a message to SNS.



Step 1: Create AWS Lambda

36. In the **AWS Management Console**, on the **Services** menu, click **Lambda**.

37. Click **Create a function**.

38. Select **Author from scratch** and configure:

- a. **Name:** Write **RemoveS3PublicAccess**.
- b. **Runtime:** Dropdown and Select **Python 3.6**.
- c. **Expand** *Change default execution role*.
 - i. **Role:** Select **Use an existing role**.
 - ii. **Existing role:** Dropdown and Select **Lambda-Config-Role**.
- d. Select **Create function**.

Note: RemoveS3PublicAccess function gets open the configuration section.

39. **Select** the **Code** section:

- a. Click on **lambda_function.py**.
 - i. **Replace** the **existing code** and **copy** the code into the **editor** from **RemoveS3PublicAccess-Lambda-Function-Code.txt** file.

Note: **Code** for **RemoveS3PublicAccess-Lambda-Function-Code.txt** is available with the Lab manual.

- b. Select **Deploy**.

40. **Select** the **Configuration** section:

- a. Select **Environment variables**.
 - i. Select **Edit**.
 - ii. Select **Add environment variables**.
 - a) **Key:** Write **TOPIC_ARN**.
 - b) **Value:** Write **Topic ARN** of **ConfigSNSTopic**.

Environment variables

You can define environment variables as key-value pairs that are accessible from your function code. These are useful to store configuration settings without the need to change function code. [Learn more](#)

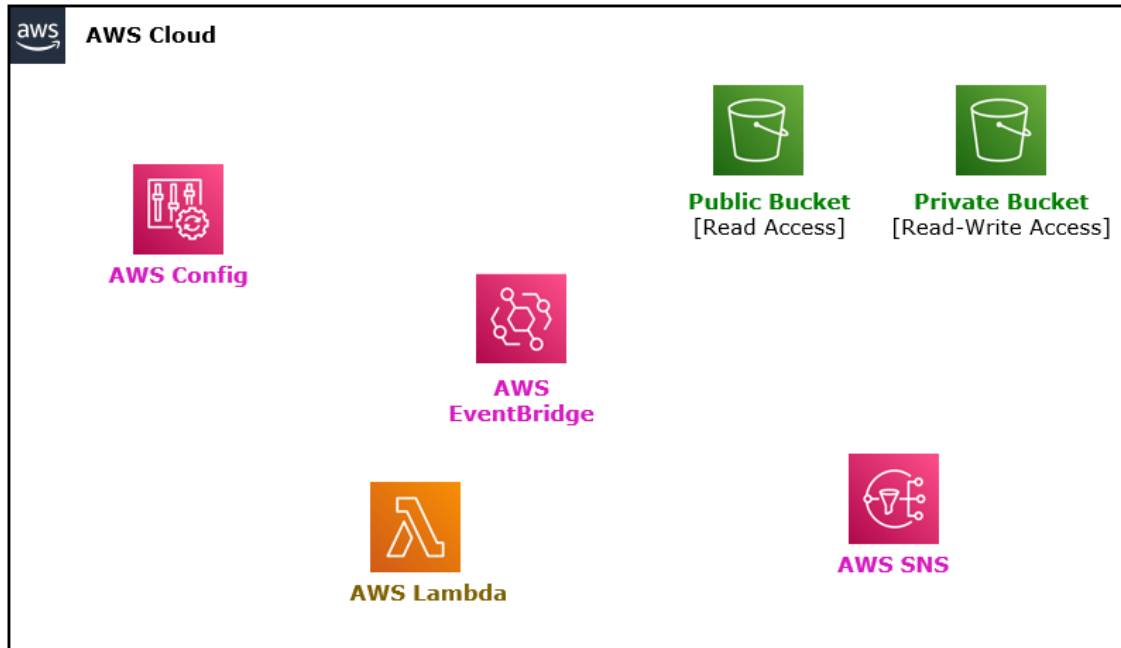
Key	Value	
TOPIC_ARN	arn:aws:sns:us-east-1:123456789012:ConfigSNSTopic	Remove

[Add environment variable](#)

- iii. Select **Save**.

Task 6: Configure AWS CloudWatch Events

In this task, you need to define how that Lambda function is going to be invoked. You are configuring the two targets: the Lambda function and an Amazon SNS topic. When a policy violation is detected, the Lambda function will fix the issue and push a message to Amazon SNS, which will send a notification to the subscribed users.



Step 1: Configure AWS EventBridge Events

41. In the **AWS Management Console**, on the **Services** menu, click **EventBridge**.
42. **Go to the left**, select **Rules**.
 - a. Select **Create Rule**.
 - b. In the **Rule details** section:
 - i. **Name**: Write **ConfigNonCompliantS3Event**.

Note: Leave other details as default.

Name

ConfigNonCompliantS3Event

Description - optional

Enter description

Event bus [Info](#)

Select the event bus this rule applies to, either the default event bus or a custom or partner event bus.

default

☒ Enable the rule on the selected event bus

Rule type [Info](#)

☒ Rule with an event pattern
A rule that runs when an event matches the defined event pattern. EventBridge sends the event to the specified target.

☐ Schedule
A rule that runs on a schedule

ii. Select **Next**.

c. In the **Event pattern** section:

i. Select **Edit pattern**.

a) **Copy** the **json** into the **editor** from **Cloud Watch Event Patternne.txt** file.

Note: **Code** for **Cloud Watch Event Patternne.txt** is available with the Lab manual.

Event pattern

Write an event pattern in JSON. You can test the event pattern against the sample event. You can also go to pre-defined pattern.

Select matching pattern

Insert

☐ Content-based filter syntax

```

1 {
2   "source": [
3     "aws.config"
4   ],
5   "detail": {
6     "requestParameters": {
7       "evaluations": {
8         "complianceType": [
9           "NON_COMPLIANT"
10        ]
11      }
12    },
13    "additionalEventData": {
14      "managedRuleIdentifier": [
15        "S3_BUCKET_PUBLIC_READ_PROHIBITED",
16        "S3_BUCKET_PUBLIC_WRITE_PROHIBITED"
17      ]
18    }
19  }
20 }

```

☒ JSON is valid

- ii. Select **Next**.
- d. In the **Target 1** section:
 - i. **Target types:** Select **AWS service**.
 - ii. **Select a target:** Dropdown and select **Lambda function**.
 - iii. **Function:** Dropdown and select **RemoveS3PublicAccess**.

Target types
Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus
☐ EventBridge API destination
☒ AWS service

Select a target [Info](#)
Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Lambda function ▼

Function
RemoveS3PublicAccess ▼

- iv. Select **Another Target**.
- e. In the **Target 2** section:
 - i. **Target types:** Select **AWS service**.
 - ii. **Select a target:** Dropdown and select **SNS Topic**.
 - iii. **Function:** Dropdown and select **ConfigSNSTopic**.

Target types
Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

☐ EventBridge event bus
☐ EventBridge API destination
☒ AWS service

Select a target [Info](#)
Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

SNS topic ▼

Topic
ConfigSNSTopic ▼

- iv. Select **Next**.

- f. In the **Configure tags** section:

Note: Leave all the details as default.

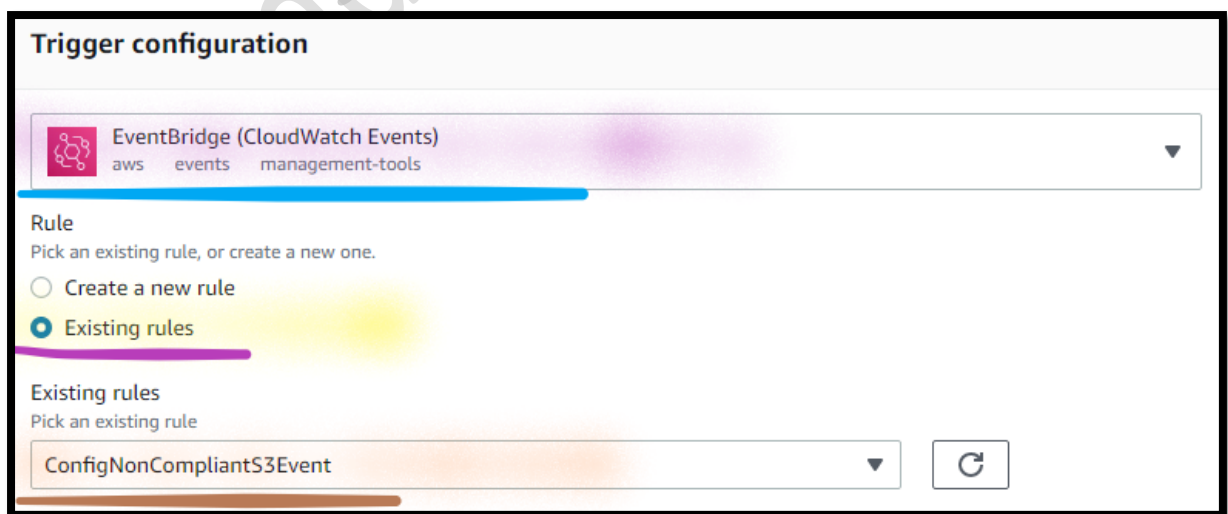
- i. Select the **Next**.
- g. In the **Review and create** section:
- i. Select the **Create rule**.

Step 2: Update the AWS Lambda Function

43. In the **AWS Management Console**, on the **Services** menu, click **Lambda**.

44. Select **function**.

- a. Open the **RemoveS3PublicAccess**.
- b. **Select** the **Configuration** section:
- i. Select **Environment variables**.
- a) Select **Triggers**.
- 1) **Trigger configuration**: Dropdown and select **EventBridge**.
- 2) **Rule**: Select **Existing rules**.
- 3) **Existing rules**: Dropdown and select **ConfigNonCompliantS3Event**.



Trigger configuration

EventBridge (CloudWatch Events)
aws events management-tools

Rule
Pick an existing rule, or create a new one.

☐ Create a new rule

☒ Existing rules

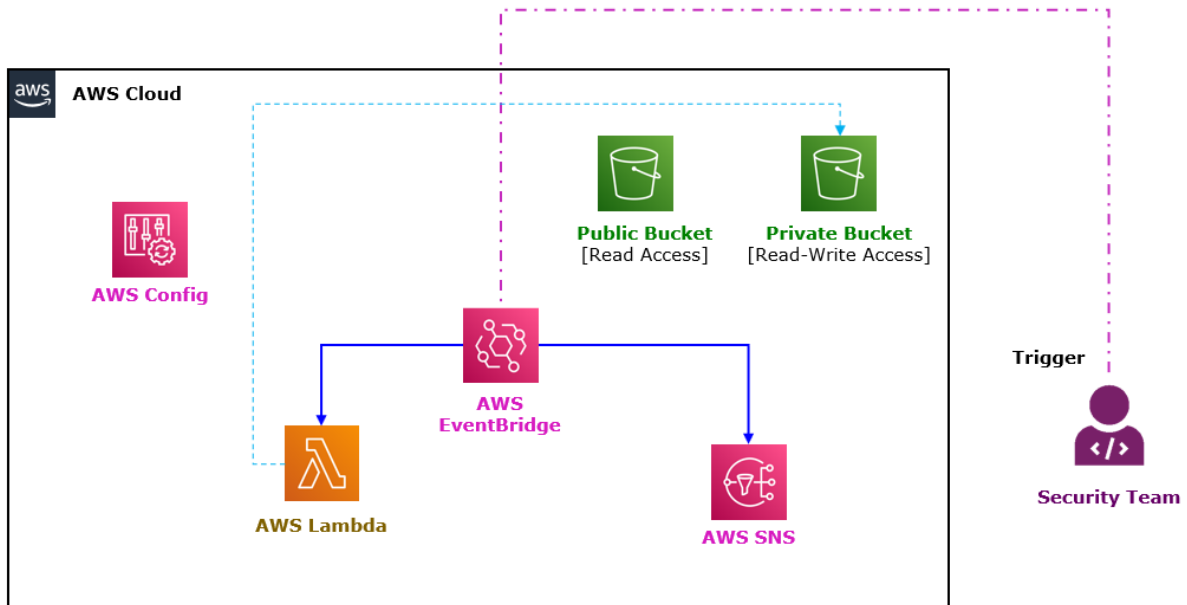
Existing rules
Pick an existing rule

ConfigNonCompliantS3Event

- 4) Select **Add**.

Task 7: Evaluate the Rule

In this task, you can test the deployment. The Lambda function begins to review your Amazon S3 buckets and automatically fix the public access.



Step 1: View the Non-Compliant Resources

45. In the **AWS Management Console**, on the **Services** menu, click **Config**.

46. **Go to the left**, select **Rules**.

a. Open the **s3-bucket-public-read-prohibited**.

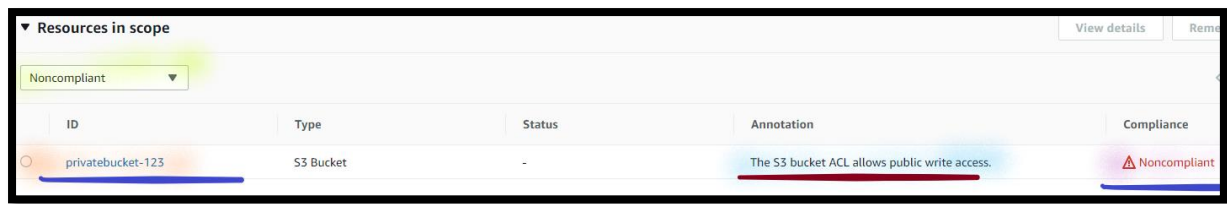
i. You can see the **Non-Compliant** resource due to **The S3 bucket ACL allows public read access**.

The screenshot shows the AWS Config console with the following table of resources in scope:

ID	Type	Status	Annotation	Compliance
publicbucket-123	S3 Bucket	-	The S3 bucket ACL allows public read access.	Noncompliant

b. Open the **s3-bucket-public-write-prohibited**.

i. You can see the **Non-Compliant** resource due to **The S3 bucket ACL allows public write access**.



The screenshot shows the 'Resources in scope' section of the AWS Config console. A dropdown menu is set to 'Noncompliant'. Below it is a table with columns: ID, Type, Status, Annotation, and Compliance. One resource is listed: 'privatebucket-123' of type 'S3 Bucket'. The status is '-', the annotation is 'The S3 bucket ACL allows public write access.', and the compliance status is 'Noncompliant' with a red warning icon.

ID	Type	Status	Annotation	Compliance
privatebucket-123	S3 Bucket	-	The S3 bucket ACL allows public write access.	Noncompliant

Step 2: Evaluate the Write-Prohibited Rule

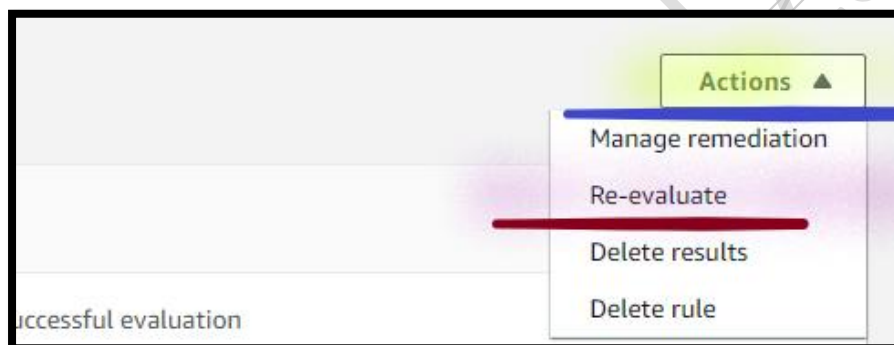
47. From the **Config** console:

48. **Go to the left**, select **Rules**.

a. Open the **s3-bucket-public-write-prohibited**.

i. Select **Actions**.

ii. Select **Re-evaluate**.



49. **Go to the left**, select **Rules**.

Note: You can see the **s3-bucket-public-write-prohibited** Rule as **Compliant**.

a. Open the **s3-bucket-public-write-prohibited**.

i. You can't see any **Non-Compliant** resource due to **Public access** removed from the **privatebucket-123**.

Step 3: Evaluate the Read-Prohibited Rule

50. From the **Config** console:

51. **Go to the left**, select **Rules**.

a. Open the **s3-bucket-public-read-prohibited**.

i. Select **Actions**.

ii. Select **Re-evaluate**.

52. **Go to the left**, select **Rules**.

Note: You can see the **s3-bucket-public-read-prohibited** Rule as **Non-Compliant**.

a. Open the **s3-bucket-public-read-prohibited**.

i. You can see **Non-Compliant** resource due to the Lambda function will treat it policy violation if **tag CanBePublic** **can't** set to **1**.

Task 8: Cleanup the Environment

Step 1: Delete the S3 Bucket

53. In the **AWS Management Console**, on the **Services** menu, click **S3**.

54. Select **Buckets**.

Delete the First bucket

55. Select **configbucket-123** bucket.

a. Select **Empty**.

b. **Type permanently delete** to **Delete all the objects**.

c. Select **Empty**.

d. Select **Exit**.

56. Select **configbucket-123** bucket.

a. Select **Delete**.

b. **Type configbucket-123** bucket name, to **Delete bucket**.

c. Select **Delete bucket**.

Delete the Second bucket

57. Select **publicbucket-123** bucket.
- Select **Empty**.
 - Type **permanently delete** to **Delete all the objects**.
 - Select **Empty**.
 - Select **Exit**.
58. Select **publicbucket-123** bucket.
- Select **Delete**.
 - Type **publicbucket-123** bucket name, to **Delete bucket**.
 - Select **Delete bucket**.

Delete the Third bucket

59. Select **privatebucket-123** bucket.
- Select **Empty**.
 - Type **permanently delete** to **Delete all the objects**.
 - Select **Empty**.
 - Select **Exit**.
60. Select **privatebucket-123** bucket.
- Select **Delete**.
 - Type **privatebucket-123** bucket name, to **Delete bucket**.
 - Select **Delete bucket**.

Step 2: Delete Lambda Function

61. In the **AWS Management Console**, on the **Services** menu, click **Lambda**.
62. Click the **Functions**.
- Select the **RemoveS3PublicAccess**.
 - Select **Actions**.
 - Select **Delete**.
 - Select **Delete**.

Step 3: Delete AWS Config Rules

63. In the **AWS Management Console**, on the **Services** menu, click **Config**.

64. **Go to the left**, select **Rules**.

Delete the First rule

a. Open the **s3-bucket-public-read-prohibited**.

- i. Select **Actions**.
- ii. Select **Delete rule**.
- iii. When prompt, type **Delete**.
- iv. Select **Delete**.

Delete the First rule

b. Open the **s3-bucket-public-write-prohibited**.

- i. Select **Actions**.
- ii. Select **Delete rule**.
- iii. When prompt, type **Delete**.
- iv. Select **Delete**.

Step 4: Delete AWS Config Recorder

65. In the **AWS Management Console**, on the **Services** menu, click **Config**.

66. **Go to the left**, select **Settings**.

- a. Select **Edit**.
- b. **Uncheck** the **Enable recording**.
- c. Select **Confirm**.
- d. Select **Ok**.

Step 5: Delete AWS CloudWatch Events

67. In the **AWS Management Console**, on the **Services** menu, click **EventBridge**.

68. **Go to the left**, select **Rules**.

a. Select **ConfigNonCompliantS3Event**.

i. Select **Delete**.

ii. Select **Delete**.

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